College of Business
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FinTech and AI in Finance

Class 2: Return Predictability and OOS Design

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From the Data Work to the Regression Work

Class 1: Build the data

- build the 240-month market and macro tables;
- define the forecast clock;
- construct momentum, volatility, and trend predictors; and
- plot the raw data and features.

Class 2: Predict and evaluate

- distinguish several meanings of return predictability;
- estimate a transparent regression forecast;
- test it only on later observations; and
- carry the frozen forecasts into Class 5 for portfolio construction.

Keep in mind

Start with the fitted relation and its statistical precision. Then ask whether the fitted model improves forecasts and investment decisions in later data.

Return Predictability Has Several Meanings

Meaning	Question	Evidence
Conditional mean	Does $E[r_t \mid x_{t-1}]$ change with the predictor?	slope, bins, fitted relation
Statistical	Is the estimated relation precise in the fitted sample?	standard error, confidence interval, p-value
Out-of-sample	Does the frozen forecast beat a simple forecast later?	MSE and OOS R^2 versus lagged MA(12)
Direction or state	Does it identify positive or high-return months?	hit rate versus the relevant base rate
Economic	Does a feasible rule improve an investment outcome?	strategy return, risk, Sharpe ratio, drawdown, exposure

Regression with AI-Assisted Coding

James Simons: Prediction as a Research Process



James (Jim) Simons
Renaissance Technologies

Research model	Left academia in 1978 to build what became Renaissance and apply mathematical modeling to traded markets.
Research team	Recruited mathematicians, physicists, and computer scientists and organized them around a shared research process.
What we take from it	Protect timing, test predictions on later data, and connect forecasts to a monitored investment rule.

Keep in mind

A regression matters only inside a process that protects timing, testing, execution, and monitoring.

Sources: Simons Foundation, biography; Simons Foundation, history. Portrait is an AI-generated editorial illustration, not a photograph.

What Regression Can Tell Us

For information x_{t-1} known before forecast month t:

$$r_t = \alpha + \beta x_{t-1} + \varepsilon_t.$$

- α is the fitted monthly market return when $x_{t-1} = 0$.
- β is the fitted change in next-month return for a one-unit change in x_{t-1}.
- ε_t is the part not summarized by this line.
- The regression describes conditional association in this sample.

Questions that still remain

Is the relation causal, stable, tradable, or useful in a genuinely new period?

A Clear Instruction Specifies the Regression

Tell the coding assistant:

1. **Question:** explore association, not claim causality.
2. **Rows:** one forecast month per row, ordered from 2006-07 to 2026-06.
3. **Target and predictor:** the return realized in month t, predicted with information known by month-end $t-1$.
4. **Model and outputs:** intercept plus named predictors; return plots, a coefficient table, fit statistics, residual checks, and saved files.
5. **Validation:** use a date split; never shuffle rows.

Explore Before You Fit

Reload the Market Feature Table

```
d = pd.read_csv(
    "market_features.csv",
    parse_dates=["forecast_month"],
).sort_values("forecast_month")
assert len(d) == 240
train = d[d["forecast_month"] < "2021-07-01"].copy()
```

Read the logic

1. parse dates;
2. sort explicitly;
3. verify 240 rows; and
4. copy the 180 pre-test months.

Keep in mind

Recheck the file at the point of use. A correct Class 1 script does not guarantee that the loaded file is correct.

Restate the Forecast Clock

At month-end $t-1$:

$$x_{t-1} \quad \longrightarrow \quad r_t = P_t/P_{t-1} - 1.$$

- `momentum_12`: 12-month price change ending at $t-1$;
- `volatility_12`: 12-month volatility ending at $t-1$;
- `ma_gap_10`: month-end price relative to its trailing 10-month average; and
- `market_return`: return realized during forecast month t.

Keep in mind

The row is labeled by the forecast month; its predictors stop one month earlier.

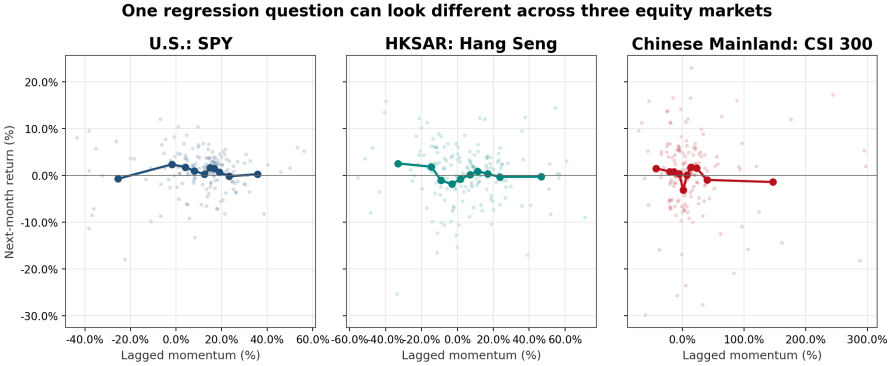
Ask Codex to Plot before Fitting

```
plt.scatter(
    train["momentum_12"],
    train["market_return"],
    s=24, alpha=0.55,
)
plt.xlabel("Lagged 12-month momentum")
plt.ylabel("Next-month market return")
```

Four choices matter

- lagged predictor on the horizontal axis;
- realized return on the vertical axis;
- transparency to reveal dense regions; and
- labels that preserve units and timing.

One Regression Question Looks Different across Markets



Keep in mind

The same lagged-momentum regression produces three noisy clouds. The binned relation and fitted slope do not travel automatically across markets.

Why Equal-Count Bins Help

Sort observations by x_{t-1}, divide them into ten groups of similar size, and compute:

$$\bar{x}_j = \frac{1}{n_j} \sum_{t \in j} x_{t-1}, \qquad \bar{r}_j = \frac{1}{n_j} \sum_{t \in j} r_t.$$

- Each point now summarizes roughly one-tenth of the sample.
- The plot reveals monotonicity, curvature, and influential tails.
- Binning is a diagnostic, not a substitute for reporting the raw data.

Ask Codex to Build Equal-Count Bins

```
train["bin"] = pd.qcut(
    train["momentum_12"].rank(
        method="first"),
    10, labels=False,
) + 1
bins = train.groupby("bin")[[
    "momentum_12",
    "market_return",
]].mean()
```

Read the logic

1. rank the predictor;
2. form ten groups of similar size;
3. average x and r within each group; and
4. plot the ten group means.

Correlation Is a Useful First Summary

For variables x and y:

$$\operatorname{corr}(x, y) = \frac{\operatorname{cov}(x, y)}{\operatorname{sd}(x)\operatorname{sd}(y)}.$$

In the 180-month pre-test sample, 2006-07–2021-06:

	Momentum	Volatility	MA gap
Next-month market return	0.02	0.12	0.05

Keep in mind

All three linear associations are small. Short-horizon returns behave this way, so read the size as information about the problem itself.

Scale and Outliers Matter

- A 0.01 change in momentum means one percentage point, not one whole unit.
- Volatility is annualized, while the target is a monthly simple return.
- A few crisis months can dominate squared-error regression.
- Clipping or deleting observations changes the estimand and must be disclosed.

Class discussion

Ask the coding assistant to list the five largest absolute returns. Should any be removed? What evidence would justify that decision?

Simple Regression

The Fitted Line

OLS chooses $\widehat{\alpha}$ and $\widehat{\beta}$ to minimize squared residuals:

$$(\widehat{\alpha}, \widehat{\beta}) = \arg\min_{a,b} \sum_{t \in \mathcal{S}} (r_t - a - b x_{t-1})^2 .$$

Then:

$$\widehat{r}_t = \widehat{\alpha} + \widehat{\beta} x_{t-1}, \qquad \widehat{\varepsilon}_t = r_t - \widehat{r}_t.$$

Keep in mind

The fitted line summarizes this sample. Whether the same line holds next year is a separate question, and one we have not asked yet.

Four Questions to Ask about a Regression

Question	Output to inspect
Direction and magnitude	slope estimate in economically meaningful units
Uncertainty	standard error and confidence interval
Fit	fitted values, residuals, and R^2
Forecast use	a method that accepts later predictor rows without refitting

Keep in mind

Decide which outputs answer your economic and forecasting question, then read only those. A full summary print is where you start looking.

Fit the Simple Regression

Estimated on the 180 pre-test months, 2006-07–2021-06:

$$r_t = \alpha + \beta \, \mathrm{Momentum}_{t-1} + \varepsilon_t.$$

Keep in mind

Write the equation before you ask for the fit. A regression output is only readable once you know which number answers which question.

Interpret the Momentum Coefficient in Units

For the 180-month pre-test sample:

$$\widehat{r}_t\,(\%) = 0.8 + 0.6\,\text{momentum}_{t-1,12}.$$

If momentum rises by 0.10, the fitted next-month return changes by:

$$0.6 \times 0.10 \approx 6.3 \text{ basis points},$$

using the unrounded coefficient for the final conversion.

Poor interpretation

"Momentum causes next month's market return to rise."

Better interpretation

Within this pre-test sample, a 10-percentage-point increase in lagged 12-month momentum is associated with a 6.3-basis-point higher fitted next-month market return.

Reading the Standard Error

For the momentum slope:

Coefficient	Estimate	Standard error	t statistic	p-value
Momentum	0.6 pp	2.1 pp	0.31	0.76

- A standard error describes how much an estimate would vary across repeated samples generated by the same process.
- The estimate is small relative to its sampling uncertainty.
- A large p-value is not proof that the true relation is exactly zero.
- A small p-value would not prove stability, causality, or economic value.
- Monthly financial residuals may still violate the simplest standard-error assumptions.

R^2 Is a Variance Summary

$$R^2 = 1 - \frac{\sum_t \widehat{\varepsilon}_t^2}{\sum_t (r_t - \overline{r})^2}.$$

For the simple momentum regression, $R^2 \approx 0.1\%$: the fitted line explains almost none of the pre-test return variation.

Keep in mind

Low R^2 is normal for short-horizon returns. It says the signal is weak relative to the noise. Whether a weak signal is still worth trading is a portfolio question, and Class 5 answers it.

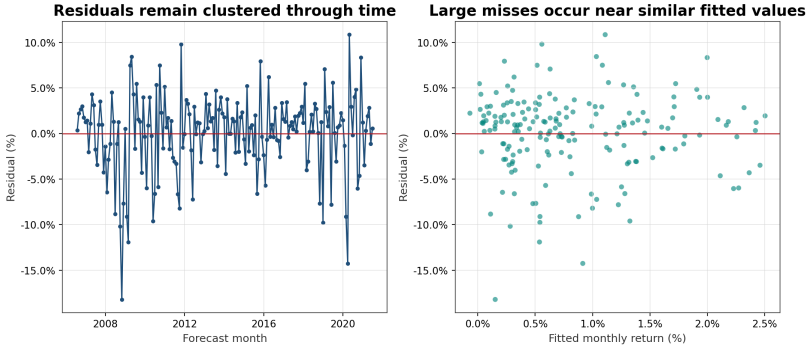
Compare the Three Simple Regressions

Predictor	Slope (pp)	Standard error (pp)	p-value	R^2
12-month momentum	0.6	2.1	0.76	0.1%
12-month volatility	8.5	5.2	0.11	1.5%
10-month MA gap	3.0	4.2	0.48	0.3%

p-values are not a predictor ranking

The three variables have different units, are correlated with one another, and were examined in the same sample.

Residuals Are Data, Not Waste



Keep in mind

The fitted values stay in a narrow range while realized errors cluster in stressful periods. Ask the AI to list the largest residual dates, but use external evidence before attaching an economic story.

Multiple-Choice Question: Interpreting a Regression Slope

A fitted regression is

$$\widehat{r} = 0.20 + 0.30m,$$

where both the return r and momentum m are measured in percentage points.

If momentum rises from 2 to 4, how much does the forecast return change?

A) It rises by 0.30 percentage points.

B) It rises by 0.80 percentage points.

C) It rises by 1.40 percentage points.

D) It rises by 0.60 percentage points.

Multiple-Choice Question: Statistical Precision and Model Fit

A return regression has a slope of 0.40, a standard error of 0.10, and an in-sample R^2 of 0.03. The reported p-value is below 0.05.

Which interpretation is justified?

A) The positive association is statistically significant, but the model explains only a small share of return variation.

B) The model must explain at least 40% of return variation.

C) The low R^2 means the slope cannot be statistically significant.

D) The significant slope establishes that the model will forecast future returns accurately.

Multiple Regression

Why Add More Than One Predictor?

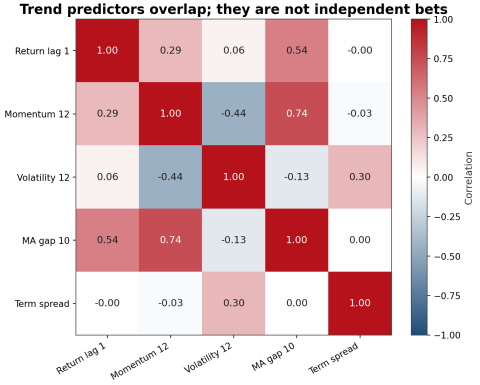
$$r_t = \alpha + \beta_1 \text{Momentum}_{t-1} + \beta_2 \text{Volatility}_{t-1} + \beta_3 \text{MA gap}_{t-1} + \varepsilon_t.$$

- A simple slope mixes the predictor's own relation with correlated variables.
- Multiple regression asks how returns vary with one predictor while holding the others fixed.
- Adding controls can change a slope's sign and magnitude.
- More predictors also increase estimation noise and researcher discretion.

Keep in mind

Putting every available predictor into one regression is the *kitchen sink* of Welch and Goyal (2008 RFS). In their data it fits the sample best and loses to the historical mean out of sample by a wide margin. Today's three-predictor model is a small kitchen sink; Class 3 keeps the sink and adds a penalty.

Our Predictors Are Strongly Related



Keep in mind

Momentum and the moving-average gap have a 0.74 pre-test correlation. Their multiple-regression slopes should not be read like independent bets.

Correlated Predictors Make Individual Slopes Unstable

- **What you see:** a coefficient changes sign or size between the simple and the multiple regression, its standard error widens, and yet R^2 barely moves.
- **Why:** when two predictors move together the data cannot say which one carries the effect, so the fitted line is well determined while the split between them is not.
- **How to check:** refit on a different subsample, or drop one of the pair, and see whether the surviving coefficients stay where they were.
- **What to do here:** report the forecast rather than the individual slope, and stop treating one coefficient as the finding.

Keep in mind

Correlated predictors are not a defect in the data. They are the normal state of financial predictors, and Class 3 answers them directly: shrink the slopes, or replace them with a few combinations.

Standardize Using Training Data Only

For each predictor j:

$$z_{j,t} = \frac{x_{j,t} - \overline{x}_{j,\text{train}}}{s_{j,\text{train}}}.$$

- One unit now means one training-sample standard deviation.
- Slopes become easier to compare across different predictor units.
- Use the same training mean and standard deviation on test observations.
- Recomputing them on the full sample leaks future distribution information.

Compare the Multiple Regression

Standardize each predictor on the training sample, $z_{t-1} = (x_{t-1} - \bar{x}_{\text{train}})/s_{\text{train}}$, then estimate

$$r_t = \alpha + \beta_1 z_{t-1}^{\text{mom}} + \beta_2 z_{t-1}^{\text{vol}} + \beta_3 z_{t-1}^{\text{gap}} + \varepsilon_t.$$

- Standardizing puts the three slopes in one unit, so their sizes can be compared with each other.
- Each $\widehat{\beta}$ now answers: how does the forecast move when this predictor rises by one training standard deviation, with the other two held fixed?

Keep in mind

This is the specification whose standardized slopes we read later. Estimating it on one sample and rescaling it on another would silently change the question.

What Changed in the Multiple Regression?

Standardized predictor	Slope	Standard error	p-value
Momentum	0.4%	0.6%	0.50
Volatility	0.7%	0.4%	0.07
MA gap	0.0%	0.5%	0.95

$$R^2_{\text{train}} = 2.2\%.$$

Keep in mind

The volatility coefficient becomes the largest standardized slope, and its uncertainty stays substantial. Conditioning on the other predictors moves the estimate, and the standard error tells you how far to trust the new position.

CSI 300 Case: Can China Macro Data Predict Next-Month Returns?

Target

$$r_t^{\mathrm{CSI300}} = \alpha + \boldsymbol{\beta}' \mathbf{x}_{t-2} + \varepsilon_t.$$

The left-hand side is the CSI 300 log return during forecast month t.

Forecast clock

- Use macro reference-month values through $t-2$.
- Estimate the fixed model on July 2006–June 2021.
- Reserve July 2021–June 2026 for a later OOS test.

Two-month availability lag; later data revisions remain, so this is not a real-time vintage test.

Four China macro predictors

- export growth, year on year;
- import growth, year on year;
- OECD composite leading-indicator gap from 100; and
- 12-month real effective exchange-rate change.

CSI 300: Traditional Predictor Significance

Evidence in the 180-month training sample

- Import growth: $\widehat{\beta} = -0.15$ percentage points per one percentage-point increase, with $p = 0.001$.
- The joint F-test has $p = 0.026$.
- Training-sample R^2 is 6.1%.

Interpretation at this stage

- The import-growth estimate is statistically distinguishable from zero, conditional on the other predictors.
- The joint test rejects the hypothesis that all four slope coefficients are zero.
- These are traditional in-sample results. They do not yet establish performance in a later period.

Keep in mind

First ask whether the fitted sample contains a precise relation. Next, test the frozen specification on later data.

Predicting Excess Stock Returns Out of Sample: Can Anything Beat the Historical Average?

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The Review of Financial Studies, Volume 21, Issue 4, July 2008, Pages 1509–1531,
https://doi.org/10.1093/rfs/hhm055
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The Review of Financial Studies, Volume 21, Issue 4, July 2008, Pages 1455–1508,
https://doi.org/10.1093/rfs/hhm014
Published: 17 March 2007

Keep Financial Time Series in Date Order

Design	What the model sees	Why it matters
Random split	a mixture of early and late months	a 2010 test month may be evaluated using a model fitted with 2025 data
Chronological split	earlier dates for fitting, later dates for evaluation	reproduces the information order faced by a real forecaster

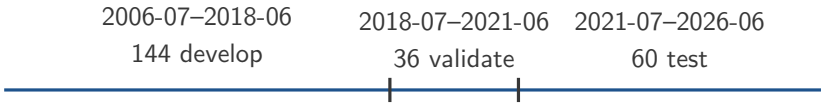
Keep rows in date order, and estimate transformations, feature choices, and coefficients before evaluating later observations.

Training, Validation, and Test Have Different Jobs

Sample	Use it to	Do not use it to
Training	estimate coefficients, means, and standard deviations	report final performance
Validation	choose predictors, transformations, or tuning settings	repeatedly rewrite the economic question
Test	evaluate the frozen design once	select the model or fix a weak result

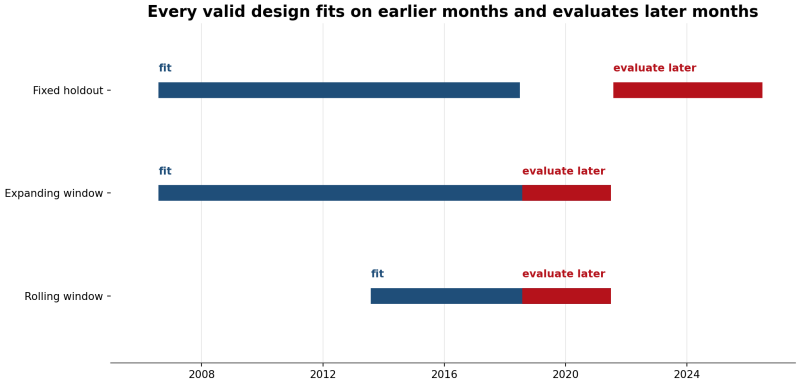
Changing a model after inspecting the test result turns that period into development data.

Use a Fixed Timeline for Today's Exercise



Develop on 144 months, select on 36 months, refit on all 180 pre-test months, and evaluate on the final 60 months. For forecast month t, MA(12) uses only $r_{t-12}, \ldots, r_{t-1}$.

Fixed, Expanding, and Rolling Tests Answer Different Questions



Keep in mind

A fixed holdout is the simplest first check. Expanding and rolling designs make more forecasts, but every valid design preserves time order and fits before evaluating.

Tell Codex to Split by Date before Fitting

```
develop = d[d["forecast_month"] < "2018-07-01"].copy()
valid = d[(d["forecast_month"] >= "2018-07-01") &
          (d["forecast_month"] < "2021-07-01")].copy()

train = d[d["forecast_month"] < "2021-07-01"].copy()
test = d[d["forecast_month"] >= "2021-07-01"].copy()
```

The four objects

- `develop`: estimate candidates;
- `valid`: choose the design;
- `train`: refit the frozen design; and
- `test`: evaluate once.

How We Will Judge a Return Forecast

No single statistic settles whether returns are predictable.

1. **Sample association:** slope, uncertainty, and in-sample R^2.
2. **Forecast error:** MSE, RMSE, and MAE in untouched observations.
3. **Relative accuracy:** improvement over a stated benchmark.
4. **Direction:** whether predicted and realized signs match, relative to the base rate.
5. **Stability:** whether results persist through time and across reasonable designs.
6. **Economic value:** whether a feasible portfolio benefits after costs, which Class 5 evaluates.

Three Forecast Criteria Stay Fixed across Classes 2–4

For T untouched test months and benchmark forecasts $\widehat{r}_t^{\,b}$:

$$\mathrm{MSE}_{\mathrm{model}} = \frac{1}{T}\sum_t (r_t - \widehat{r}_t)^2,$$

$$R^2_{\mathrm{OOS}} = 1 - \frac{\sum_t (r_t - \widehat{r}_t)^2}{\sum_t (r_t - \widehat{r}_t^{\,b})^2},$$

$$\mathrm{HitRate} = \frac{1}{T}\sum_t \mathrm{I}\{\mathrm{sign}(\widehat{r}_t) = \mathrm{sign}(r_t)\}\,.$$

- Lower MSE means smaller forecast errors; positive OOS R^2 means beating the stated benchmark.
- Directional accuracy must beat the positive-return base rate, not an automatic 50% benchmark.
- A model is selected on validation data; these criteria are reported once on the untouched test.

The OOS R^2 Benchmark Must Match the Prediction Target

Prediction target	Benchmark forecast	Economic meaning
Monthly market return	$\widehat{r}^b_{m,t} = \frac{1}{12}\sum_{j=1}^{12} r_{m,t-j}$	MA(12) is a feasible forecast formed from the 12 market returns known before month t.
Weekly stock active return	$\widehat{a}^b_{i,t} = 0$	Zero says stock i is expected to match its local market; Classes 3–4 ask whether characteristics improve on that neutral forecast.

Keep in mind

OOS R^2 has no target-free meaning: always state the target, benchmark forecast, and information date together.

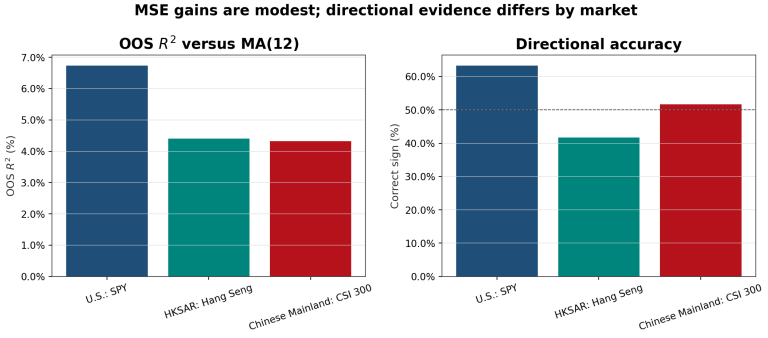
Let Codex Generate the Frozen Test Forecasts

```
test_z = test.copy()
test_z[["momentum_z", "volatility_z",
        "ma_gap_z"]] = (
    test[predictors] - mu
) / sd
test["prediction"] = fit3.predict(test_z)
```

What must stay frozen

- predictor names;
- training means and standard deviations;
- fitted coefficients; and
- test dates and row order.

Compare with Lagged MA(12) in Every Market



Keep in mind

The common OLS specification has modestly positive OOS R^2 versus MA(12) in all three markets. This loss comparison does not replace the directional base rate or an investment benchmark.

SPY Result Walkthrough: Direction Needs a Baseline

Using the hit rate defined above, the test sample gives:

Forecast	Directional hit rate
Three-predictor regression	63.3%
Lagged MA(12) forecast	61.7%
Always-positive forecast	63.3%

Keep in mind

MA(12) benchmarks MSE. Sign accuracy instead competes with the majority sign: positive returns occur in 38 of 60 months, or 63.3%.

SPY is an adjusted total-return series and includes distributions. The Hang Seng and CSI 300 series are price indices, so the U.S. column carries dividend income that the other two do not.

Hang Seng and CSI 300: Direction against Their Own Base Rates

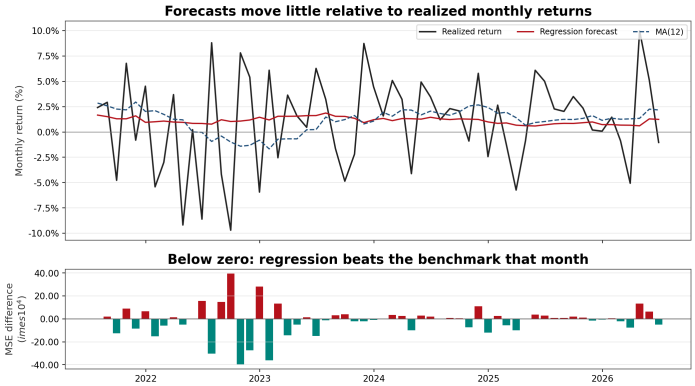
The same three-predictor regression, scored on the same 60 test months:

Market	OOS R^2 vs MA(12)	Hit rate	Positive-month base rate
Hang Seng	4.4%	41.7%	46.7%
CSI 300	4.3%	51.7%	48.3%

Keep in mind

The Hang Seng forecast beats MA(12) on MSE yet calls the sign right in 25 of 60 months, fewer than the 28 positive months, so an always-positive guess would have scored higher. The CSI 300 forecast clears its base rate by two months, and SPY only matched its own. A positive OOS R^2 and a useful direction call are different claims.

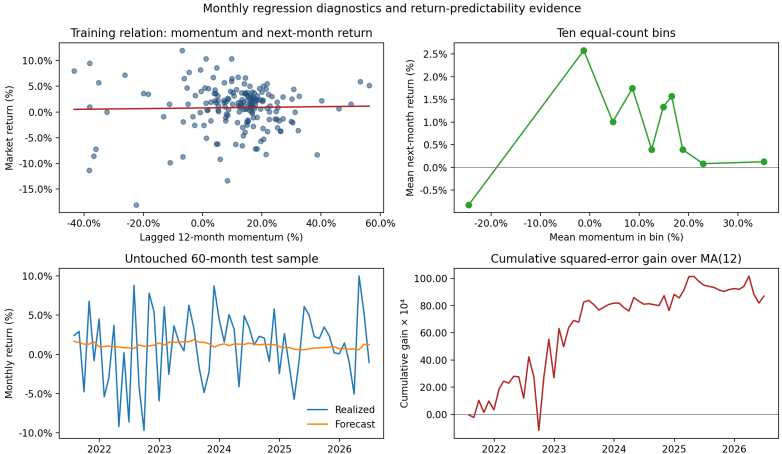
SPY Result Walkthrough: Forecasts Move Little



Keep in mind

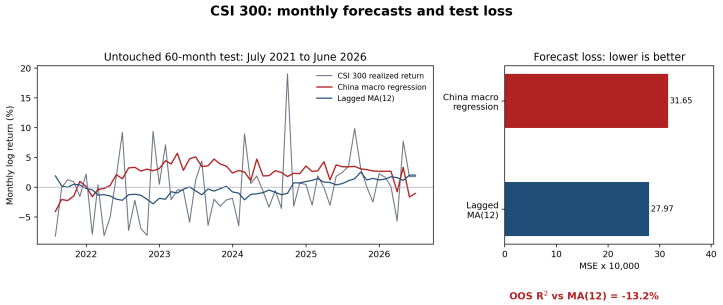
Some months favor the regression and others favor MA(12). Across all 60 test months, the regression's OOS R^2 versus MA(12) is 6.6%.

SPY Result Walkthrough: Read the Full Dashboard



Read clockwise: weak fitted relation, unstable bins, small forecasts, and a modest cumulative MSE gain over lagged MA(12).

CSI 300 Macro Model: OOS Evaluation against Lagged MA(12)



Predictors: export growth, import growth, OECD leading indicator, and real effective exchange rate; each lagged two months.

Keep in mind

On the untouched test, OOS $R^2 = -13.2\%$ versus MA(12), and the hit rate is 48.3% versus 56.7%. The significant training-sample coefficient did not improve later forecasts.

When the Regression Output Looks Wrong

Use this order:

1. print the exact columns and first five regression rows;
2. confirm target and predictor dates by hand for three rows;
3. check missing values, duplicates, units, and extreme observations;
4. fit the simplest one-predictor model;
5. compare two Python implementations or compute the slope by formula;
6. only then add transformations or more predictors.

Start with dates, rows, and units before changing the model.

Multiple-Choice Question: Would You Use This Forecast?

A volatility regression has the highest in-sample R^2 among three predictors and a plausible economic story. Which evidence most directly supports using it for real-time market-return prediction?

A) Its in-sample coefficient is positive.

B) Its residuals average to approximately zero in the fitted sample.

C) Its later forecasts deliver positive OOS R^2 relative to the lagged MA(12) benchmark using information available at each forecast date.

D) Its predictor has the largest numerical scale.

Discuss

Why do in-sample fit and a plausible story answer a different question from real-time forecast performance?

In-Class Check: From Plot to Forecast

Discuss one candidate predictor using the following evidence:

- one scatter plot and one binned-mean plot;
- one coefficient table for simple and multiple regression;
- one return-predictability scorecard compared with the benchmark;
- one explanation of why forecast accuracy alone does not establish investment value;
- a short interpretation in plain language.

Which piece of evidence would make you change your conclusion?

What to Remember

1. Plot first and read coefficients in their economic units.
2. Match the benchmark to the question: lagged MA(12) for market-return forecasts and the positive-return base rate for direction.
3. Choose with validation data, evaluate later data once, and carry the frozen forecasts into Class 5 for portfolio construction and comparison with buy-and-hold.